SakhiLine — Project Brief & Architecture

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SakhiLine is a voice assistant for ASHA health workers, built for Gnani AI's Great Indian AI Internship Challenge, with submissions due Nov 10, 2026.

What we're building

An ASHA worker taps one button after a home visit and talks; SakhiLine fills the visit form, she confirms it, and it goes to her supervisor for approval.

She speaks the way she naturally does, in her own language mixed with English medical terms: *"Sunita ka BP check kiya, Hb 9 tha, TT ka doosra injection due hai agle hafte."* Before she starts, the assistant pulls the beneficiary's history, so she only adds what is new.

The assistant works through three channels, one per network situation:

Channel	When it is used	What she gets
App, live push-to-talk	Mobile data works	Form fills on screen as she speaks; she taps any field to fix it
App, record and sync later	Weak or no data, not urgent	Recording queued on the phone, processed when the signal returns, draft sent back for her to confirm
Phone call	No data, urgent	Voice-only flow over a normal call, with a spoken read-back to confirm

Two more pieces complete the product:

- **Supervisor review.** Submitted records land in an approval queue for her ANM or block health officer, who approves them or sends comments back.
- **Built-in stress tester.** SakhiLine generates hundreds of synthetic noisy calls, runs them through its own pipeline, and publishes accuracy by language and noise level. Every accuracy claim in the demo comes with a measured number.

Why we're building it

ASHA workers lose hours to paperwork after every round of home visits, and voice removes most of it.

- **The problem.** After visits, ASHAs fill registers and apps by hand, often in English-heavy forms, then chase supervisors and mothers separately. Data entry competes with the visits themselves.
- **The users.** ASHA workers (primary), their supervisors such as the ANM or block health officer (reviewers). Rural health workers are named in the contest's Real Deal award brief.
- **Why voice.** ASHAs already describe visits out loud, in their own language, mixing in English medical terms. Speaking is faster than typing on a phone and works for workers less comfortable with English forms.
- **Why Gnani.** Gnani's models are trained on Indian telephonic audio, code-mixed speech and Indian languages. That is exactly this user's input: noisy, multilingual, often over a weak phone line.
- **Why it wins the contest.** It is easy to explain in a 60-second video, emotional enough to be shared, and backed by measured accuracy numbers instead of claims.

Constraints

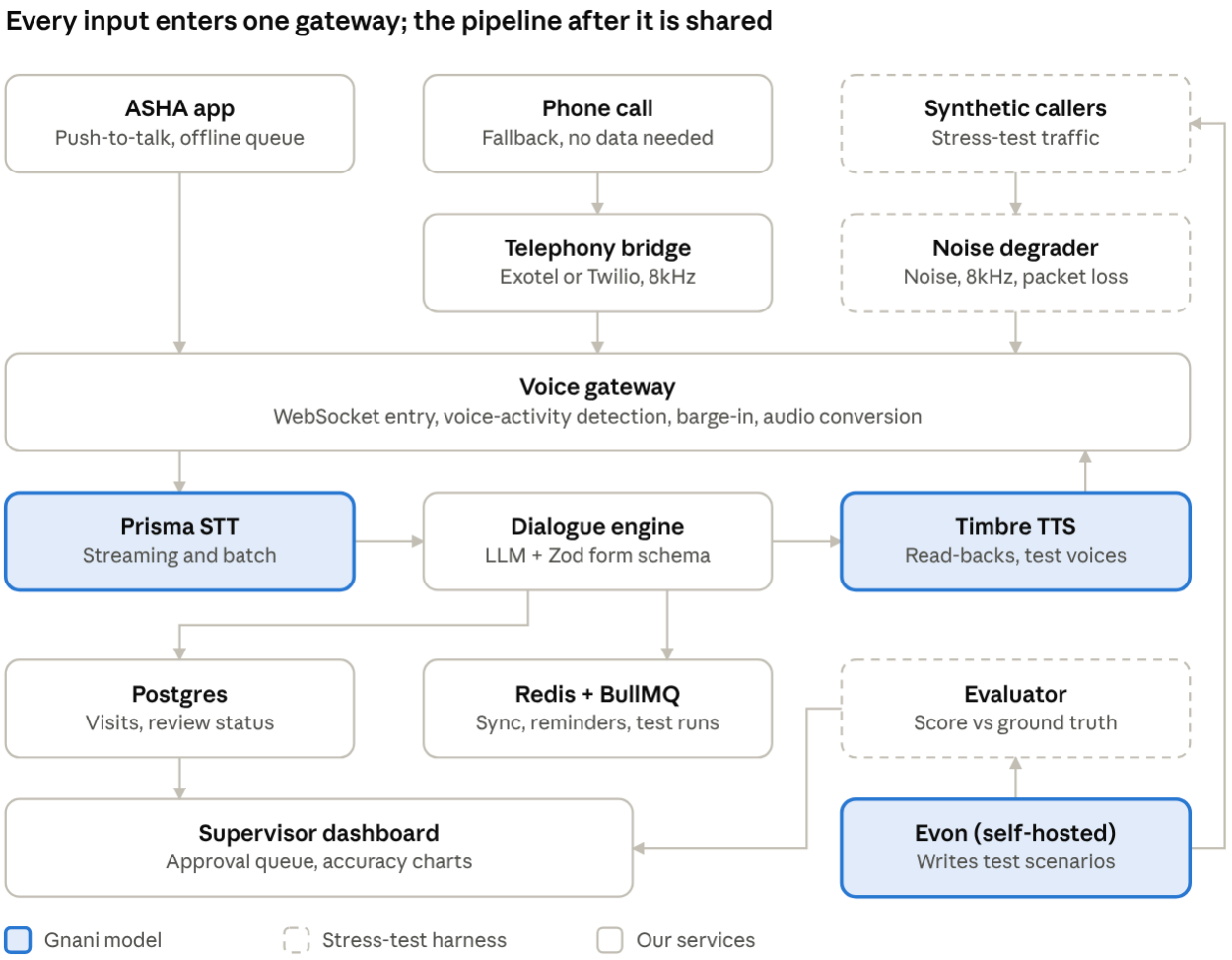
The hardest constraints are the 23-day window, synthetic-data-only rules, and Evon not being on Gnani's API.

Constraint	Detail	What it means for us
Deadline	Submissions close 11:59 PM IST, Nov 10, 2026	About 23 days of build time from Oct 18
Team	Exactly two people	Clear ownership split; each member's contribution is written into the submission
Gnani tech	Must use at least one Gnani model or API, named in the submission	Prisma and Timbre are core; Evon is used offline
Evon access	Not on the API platform; open weights by request on Hugging Face	Self-host on a rented GPU, used only for offline batch jobs
Credits	5,000 programme credits per registration	Track usage; do not load-test the live APIs on credits
Synthetic data	No real phone numbers, records, Aadhaar, PAN or recorded calls in the project or demo	All beneficiaries, readings and calls are invented, and the demo says so

Constraint	Detail	What it means for us
Agent builder	Out of scope; only the API platform counts	We build our own gateway and orchestration
Public demo	Post on LinkedIn and/or X with #GnaniAI and #GreatIndianAIInternshipChallenge before submitting	Shortlisting is ranked by verified engagement, so the post must go live early
Authentic engagement	No bought engagement, bots, pods or engagement swaps	Growth comes only from genuine reshares
Medical safety	— (our own rule)	SakhiLine logs and reminds; it never diagnoses or recommends treatment
Network	Rural data coverage is patchy	Offline queue and call fallback are part of the design, not extras

Architecture

The app, the phone call and the synthetic test traffic all enter one voice gateway, so the Prisma → dialogue engine → Timbre pipeline is built and tested once.



SakhiLine architecture · 3 inputs, 1 gateway, shared pipeline

Evon feeds the synthetic callers and the evaluator offline; confirmed records land in Postgres and surface on the supervisor dashboard.

How data flows

Every input, real or synthetic, enters through one voice gateway, so all five flows share the same pipeline after it.

1. Live app (primary)

1. She taps the button and names the beneficiary. The app fetches the beneficiary's history from the backend and shows it.
2. The app streams 16kHz microphone audio over a WebSocket to the voice gateway.
3. The gateway streams it into Prisma; partial transcripts appear on screen as she speaks.
4. The dialogue engine (an LLM with function calling) turns the transcript into form fields, validated against a strict schema (Zod). Missing or unclear fields get a follow-up question, never a guess.

5. Fields fill on screen live. Timbre speaks short prompts or the read-back if she wants audio.
6. She confirms or taps to fix a field, then submits. The record goes to Postgres with status *pending review*.

Target: under about 1.2 seconds from when she stops talking to the next prompt, which needs streaming at every stage.

2. Offline record and sync

1. With weak or no data, the app records audio locally and queues it in the phone's browser storage.
2. When the signal returns, the app uploads the queued recordings.
3. A sync worker runs each one through the same pipeline as a batch job.
4. The draft form comes back to the app; she reviews, fixes and submits as in flow 1.

3. Phone-call fallback

1. She calls a number. The telephony provider (Exotel or Twilio) streams 8kHz call audio to the gateway.
2. Same Prisma → engine → Timbre loop, voice-only, with a spoken read-back: *"Sunita, Hb 9, TT second dose next Tuesday — sahi hai?"*
3. On confirmation the record is saved as *pending review*, and she can check it in the app later.

4. Supervisor review

1. Submitted records appear in the supervisor's approval queue on the dashboard.
2. Rule-based flags, with thresholds set by the health department, push certain records to the top of the queue. The AI never decides what is risky.
3. The supervisor approves, or sends a comment back to the ASHA, who fixes and resubmits.

5. Stress test (the twist)

1. Evon, self-hosted and offline, writes hundreds of synthetic visit scenarios: a spoken script per scenario plus the correct record as ground-truth JSON.
2. Timbre voices each script as an ASHA.
3. A noise degrader adds village noise, downsamples to 8kHz, applies phone-codec compression and drops packets, at controlled levels.

4. The audio enters the same gateway as real traffic, through a simulated-caller WebSocket.
5. The evaluator compares each extracted record with its ground truth field by field and reports accuracy by language, code-mixing level and noise level, plus latency percentiles.

Where the Gnani models are used

All three Gnani models have a real job: Prisma hears, Timbre speaks and generates test voices, and Evon writes the test scenarios offline.

Model	What it is	Where SakhiLine uses it	Mode
Prisma v2.5	Speech-to-text, trained on Indian telephonic audio, code-mixed speech	Live app transcription (flow 1)	Real-time streaming
		Phone-call fallback on 8kHz audio (flow 3)	Real-time streaming
		Offline recordings uploaded later (flow 2)	Batch
		Every stress-test call (flow 5)	Streaming, through the same gateway
Timbre v2.5	Text-to-speech for Indian languages, streams over WebSocket and SSE	Read-backs and prompts in the app and on calls	Real-time streaming
		Voicing synthetic ASHA callers for the stress test	Batch
Evon v3.3	30B mixture-of-experts LLM, open weights	Writing synthetic visit scenarios and ground-truth records	Offline batch, self-hosted
		Judging free-text fields during evaluation	Offline batch, self-hosted

The real-time conversation logic runs on whichever hosted LLM is fastest, because Evon is not on Gnani's API and self-hosting it for live calls would add latency and GPU cost.

The stress test uses Gnani's text-to-speech to generate test data for Gnani's speech-to-text, so the system measures itself. That is the Jugaad Genius angle and the strongest interview talking point.

Tech stack

The backend stays on the team's existing Node.js and TypeScript stack; only the audio work and Evon hosting add Python.

Layer	Choice	Why
ASHA app	React PWA (installable on Android), push-to-talk over WebSocket, browser storage for the offline queue	One codebase, no app-store step, works offline
Supervisor dashboard	React	Same frontend stack as the app
Voice gateway	Node.js + TypeScript WebSocket server; voice-activity detection, barge-in, audio conversion	One entry point for app, call and test traffic
Dialogue engine	Node.js + TypeScript; hosted LLM with function calling; Zod schemas	Strict, validated form filling
Data	PostgreSQL + Prisma ORM	Beneficiaries, visits, review status
Jobs	Redis + BullMQ	Offline sync jobs, reminder calls, stress-test runs
Telephony	Exotel or Twilio media streams	Call fallback
Noise degrader	Python worker (numpy, librosa or ffmpeg)	Noise mixing, 8kHz downsampling, codec, packet loss
Evon hosting	vLLM on a rented GPU, switched on only for batch runs	Keeps GPU cost to the hours we use it
Evaluator	Node.js or Python job writing results to Postgres	Accuracy charts on the dashboard

Award mapping

SakhiLine is in contention for 8 of the 10 awards, but a team can win only one, so Real Deal and Noise Canceller are the primary targets.

Award	Feature that earns it	Priority
Real Deal	Rural health workers are named in the brief; visible time saved per ASHA	Primary
Noise Canceller	Measured accuracy on noisy, phone-degraded audio from the stress test	Primary
Main Character	Full flow in one regional language that Prisma and Timbre support	Secondary
Mixed Vibes	ASHAs mix their language with English medical terms (BP, Hb, TT)	Secondary
Jugaad Genius	Gnani's text-to-speech generating test data for Gnani's speech-to-text	Secondary
Two-Person Army	Clean two-person split, recorded in the submission	Automatic entry
Demo Day Drop	60-second video: village, voice, form fills, supervisor approves, accuracy chart	Secondary
Crowd Favourite	Shareable, emotional story; same engagement that drives shortlisting	Secondary
Solo Ninja	Not eligible; we are a team of two	—
Small Town, Big Build	Depends on whether our college or city is Tier 2/3	Check

Scope and open questions

The full design is too much for two people in 23 days, so build in this order and cut from the bottom.

Must ship

- [ ] App with live push-to-talk, beneficiary history fetch, and live form filling
- [ ] Supervisor approval queue on the dashboard
- [ ] Stress-test harness and the accuracy-by-noise-level chart
- [ ] 60-second demo video, and the LinkedIn/X post live well before Nov 10

Should ship

- [ ] Phone-call fallback (keeps the Noise Canceller story)

Nice to have

- [ ] Offline record-and-sync
- [ ] Rule-based supervisor flags
- [ ] Outbound reminder calls to mothers (otherwise shown as a roadmap item)

Verify in Gnani's docs before writing code

- [ ] Audio format Prisma's streaming endpoint expects (sample rate, encoding)
- [ ] Timbre's output audio format and streaming behaviour
- [ ] Which languages Prisma and Timbre each support, to pick the regional language
- [ ] Whether Prisma has a batch endpoint for the offline sync path
- [ ] How far 5,000 credits go for streaming STT and TTS